Smoothing Synchronized Power Swings in HPC Clusters

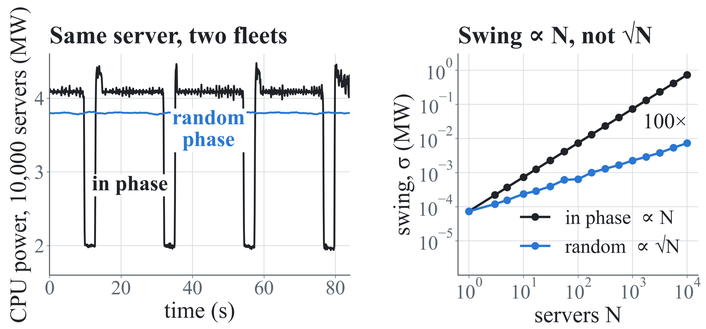
Using Software: Measurement, Prediction, and Grid Simulation

Daniel Lee[1], Emily Doyle[2], Alexander Johnson[3]

[1]Grinnell College | [2]Embry-Riddle Aeronautical University | [3]Georgia Institute of Technology

INTRODUCTION

Bulk-synchronous AI training: nodes compute, then communicate gradients to adjust weights. Thousands of servers run in lockstep, leading to huge power swings.



One measured server trace copied 10,000 times: random phases cancel to a √N ripple; in-phase copies swing 100× harder.

METHODS

M1 A live power monitor for one server

RAPL CPU power 10 Hz, C sampler
Linux /proc 1 Hz: run queue, I/O, ctx switches
Redfish BMC every 10 s: 69 fields
Rack PDU ~1.8 Hz: real power

→ **InfluxDB** Data storage → **Grafana** live dashboards, 5 s refresh

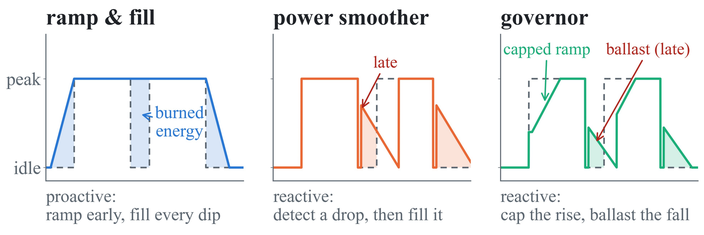


Live: three independent power telemetry monitors and CPU utilization on one Grafana dashboard. Whole team can see server activity in real time.

Cost of Running: 2.19% of one core (0.017% of the server); Running sampling code as a daemon is +0.15 W.

Correct by construction: power is reported only when the RAPL counter advances, removing the kilowatt "spikes" when polling the hardware too fast.

M2 Three software mechanisms for smoothing power



Experiment: 3 workloads (HPL, AI inference, step) × 4 conditions × 4 repeats = 48 traces on a quiesced dual Xeon Gold 6338.

M3 Grid model and evaluation criteria

1 server RAPL trace → **× 10,000** in phase, +PUE → **UPS** 15 s low-pass → **Phasor grid** 50 MW, H = 2.5 s → **Scorecard** metrics below

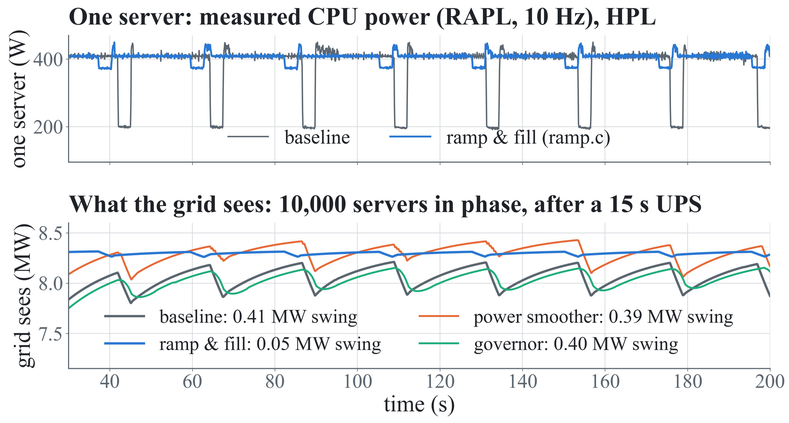
$$2H/\omega_0 \cdot d\Delta\omega/dt = P_m - P_e - D\Delta\omega \quad \textit{(swing equation)}$$

Metric	What it captures	Limit / standard
CV, peak-to-mean	volatility; worst excursion of PCC power	lower is flatter
Voltage sag	PCC voltage dips and their depth	0.95 pu (IEEE 1159)
Frequency nadir	lowest frequency after a load swing	59.3 Hz relay (NERC BAL-003)
ROCOF	how fast frequency moves	Hz/s (no fixed limit)
Ramp rate (RREI, NRS)	MW/min above the limit, annualized	20 MW/min
Runtime, energy, Gflop/s	what smoothing costs	vs. own baseline

Experiment Design: score 13 metrics for every run against its own workload's baseline. Repeat each run four times. Simulate a grid of 10,000 identical servers synchronously running to evaluate risk of damage to the grid. All metrics were custom-made, inspired by IEEE and NERC safety standards for industrial use.

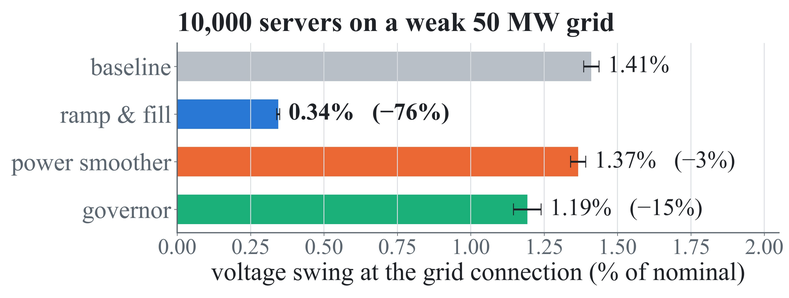
RESULTS

R1 Ramp & fill flattens power best



Top: one server, measured. Bottom: the same traces after fleet aggregation and the UPS, which acts as a 15 s low-pass filter: fast fluctuations never reach the grid, but the slow duty-cycle effects do.

R2 Grid stress: voltage swings

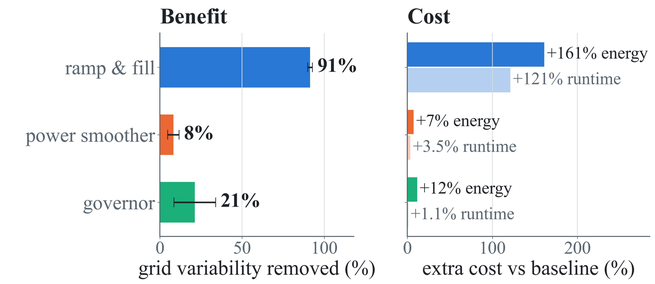


Peak-to-trough voltage at the grid connection during steady running; mean of 3 workloads ± run-to-run SD, n = 4. For scale: unsmoothed, voltage sags below the 0.95 pu IEEE limit near 10,000 servers.

R3 The cost of smoothing: runtime and energy

91%	**8%**	**21%**
ramp & fill	**power smoother**	**governor**
less grid variability (±1%)	less grid variability (±4%)	less grid variability (±13%)

Reduction in grid-facing variability (CV of facility power): mean of 3 workloads ± run-to-run SD, n = 4 repeats.



Benefit (left) against what it costs the server in energy and runtime (right).

R4 Forecasting power swings: not yet possible

Smoothing needs warning, so we tried to forecast power, then to detect it early:

Approach	What happened
Regression (OLS), 5 s ahead	caught 19 of 1,735 spikes
Random forest + Kalman filter	PR-AUC 0.68, but flags 10–20 s late
Run-queue jerk, Hawkes process	real precursor (d = 1.5), branching 0.84–0.96
CPU-utilization edge detector	best: one 10 Hz sample, 5 alarms/hr

0.23 s late median delay of the best detector on an independent workload: it fires after the transition, not before (recall 33%, precision 19%).

CONCLUSION

Voltage is the problem: in the worst case, where all servers run in lockstep, the first damage to the grid is when voltage drops too low. Voltage sags are the bottleneck for synchronous jobs.

Proactive strategy works: ramp & fill significantly flattens power and voltage swings for a premium cost. Worth it for long jobs that maintain high power, which are relevant to the NSA.

Reacting is late but viable: power swings can't be predicted yet, but the governor still trims ~20% of variation for minimal cost. Improved telemetry would only make this method better.

Deliverables: a live monitor that costs 0.017% of the server and a novel grid-risk evaluation framework, shared with NSA teams to continue power smoothing efforts in HPC clusters. NSA professionals shared our telemetry pipeline with AMD to inspire innovation in faster, more reliable power-reporting hardware.

Code · data · technical reports

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github.com/leedanie2/Collecting-and-Adjusting-Server-Power

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